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United States Patent	12398502
Kind Code	B2
Date of Patent	August 26, 2025
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Distributor, water intake system, and laundry treating device

Abstract

A distributor, a water intake system, and a laundry treating device. The distributor includes a base, the base including a water accommodating portion which is used to define a water flow boundary, and an insertion portion being formed at an upper end of the water accommodating portion; a cover plate, a fitting slot being formed in the cover plate, and the insertion portion being limited in the fitting slot; and a fastener, the fastener connecting the cover plate to the base in a fastening manner, and to form sealing faces, which are in close contact, between the insertion portion and the fitting slot. The base and the cover plate of the distributor are connected by means of the fastener. Under the fastening action of the fastener.

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Appl. No.:	18/025907
Filed (or PCT Filed):	November 30, 2020
PCT No.:	PCT/CN2020/132677
PCT Pub. No.:	WO2022/052330
PCT Pub. Date:	March 17, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240003075 A1	Jan. 04, 2024

Foreign Application Priority Data

CN

202010955169.8

Sep. 11, 2020

Publication Classification

Int. Cl.: **D06F39/08** (20060101); **D06F39/02** (20060101)

U.S. Cl.:

CPC **D06F39/088** (20130101); **D06F39/028** (20130101);

Field of Classification Search

USPC: None

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS

(1) The present disclosure is a national phase application of International Application No. PCT/CN2020/132677, filed on Nov. 30, 2020, which claims priority to Chinese Patent Application No. 202010955169.8, filed Sep. 11, 2020, the entireties of which are herein incorporated by reference.

FIELD

(2) The present disclosure relates to the field of household appliances, and in particular to a

dispenser, a water inlet system and a laundry treating device.

BACKGROUND

(3) This section provides only background information relevant to the present disclosure, and is not necessarily the related art.

(4) A water inlet system of a laundry treating device, such as a washing machine, a washing and drying machine, etc., is generally used to distribute the public tap water, and configured to allow the tap water to flow into different detergent dispensers, and allow the washing agents in the different detergent dispensers to be flushed into a washing drum of the laundry treating device. A dispenser is one of the main devices in the water inlet system, and the process for distributing the tap water is realized through the dispenser. In the related art, the dispenser is mainly composed of a base and a cover plate. The cover plate and the base are connected with each other and collectively define a flow channel for passage of water flow. During the passage of water flow, the flow channel is subjected to a level of water pressure. In order to ensure the sealing performance of the flow channel, the cover plate and the base are usually connected with each other by welding, resulting in higher cost and lower production efficiency.

SUMMARY

(5) The present disclosure aims at least to solve the problem that the existing dispenser being assembled by welding results in higher cost and lower production efficiency. The aim is achieved by the embodiments as follows.

(6) An embodiment of the present disclosure proposes a dispenser. The dispenser includes: a base, in which the base includes a water storing part configured to define a water flow boundary, an insertion part being formed at an upper end of the water storing part; a cover plate, in which a fitting slot is formed on the cover plate, the insertion part being limited in the fitting slot; and a fastener, in which the fastener fastens the cover plate to the base to enable sealing surfaces to be formed between the insertion part and the fitting slot, the sealing surfaces being in close contact with each other.

(7) According to the dispenser in the embodiment of the present disclosure, the base of the dispenser is connected to the cover plate of the dispenser through the fastener, and the water storing part on the base is configured to define the water flow boundary. In a case that the base is connected to the cover plate, the water storing part and the cover plate collectively define a water flow channel. Meanwhile, under the fastening action of the fastener, the insertion part formed at the upper end of the water storing part fits tightly into the fitting slot formed on the cover plate, and sealing surfaces, which are in close contact with each other, are formed between the insertion part and the fitting slot, to guarantee the sealing performance of the water flow channel. Since the base is connected to the cover plate by means of locking through the fastener, and meanwhile the channel is sealed, the cost can be reduced, and the production efficiency is relatively high compared to the welding method used in the related art.

(8) In addition, the dispenser in the embodiment of the present disclosure may further include the additional embodiments as follows.

(9) In some embodiments of the present disclosure, a bottom portion of the fitting slot is provided with a narrowed section. Under a fastening action of the fastener, an end of the insertion part abuts tightly against inner walls of the narrowed section to form a first sealing surface and a second sealing surface.

(10) In some embodiments of the present disclosure, a protrusion is formed at the bottom portion of the fitting slot. Under the fastening action of the fastener, the end of the insertion part abuts tightly against the protrusion to form a third sealing surface.

(11) In some embodiments of the present disclosure, a first inclined surface is formed on the insertion part, and a second inclined surface is formed on an inner wall of the fitting slot. Under the fastening action of the fastener, the first inclined surface abuts tightly against the second inclined surface to form a fourth sealing surface.

(12) In some embodiments of the present disclosure, the water storing part includes a flow channel assembly, and a spray chamber connected to the flow channel assembly. Water outlet holes are formed at a bottom part of the spray chamber.

(13) In some embodiments of the present disclosure, the flow channel assembly includes: a X-shaped flow channel, in which the X-shaped flow channel includes a first flow channel and a second flow channel intersecting with the first flow channel, in which the first flow channel includes a first water inlet section and a first water outlet section, and the second flow channel includes a second water inlet section and a second water outlet section; and a third flow channel, in which the third flow channel is arranged between the first water outlet section and the second water outlet section, the third flow channel is in communication with the X-shaped flow channel, an inlet of the third flow channel is arranged at an intersection of the first flow channel with the second flow channel, and the spray chamber is in communication with the third flow channel.

(14) In some embodiments of the present disclosure, the dispenser further includes: a first water inlet joint arranged at an inlet of the first flow channel; a second water inlet joint arranged at an inlet of the second flow channel; a first water outlet joint arranged at an outlet of the first flow channel; and a second water outlet joint arranged at an outlet of the second flow channel.

(15) In some embodiments of the present disclosure, the dispenser further includes a hot water pipe joint arranged on the cover plate. The hot water pipe joint is in communication with the spray chamber.

(16) In some embodiments of the present disclosure, the base further includes a receiving tray. The receiving tray is arranged around the flow channel assembly, and the receiving tray is provided with water leakage holes.

(17) In some embodiments of the present disclosure, the dispenser further includes a snap assembly arranged between the base and the cover plate.

(18) An embodiment of the present disclosure proposes a water inlet system, which includes the dispenser in any one of the above embodiments.

(19) According to the water inlet system in the embodiment of the present disclosure, the base of the dispenser of the water inlet system is connected to the cover plate of the dispenser of the water inlet system through the fastener, and the water storing part on the base is configured to define the water flow boundary. In a case that the base is connected to the cover plate, the water storing part and the cover plate collectively define a water flow channel. Meanwhile, under the fastening action of the fastener, the insertion part formed at the upper end of the water storing part fits tightly into the fitting slot formed on the cover plate, and sealing surfaces, which are in close contact with each other, are formed between the insertion part and the fitting slot, to guarantee the sealing performance of the water flow channel. Since the base is connected to the cover plate by means of locking through the fastener, and meanwhile the channel is sealed, the cost can be reduced, and the production efficiency is relatively high compared to the welding method used in the related art.

(20) An embodiment of the present disclosure proposes a laundry treating device, which includes the water inlet system in any one of the above embodiments.

(21) According to the laundry treating device in the embodiment of the present disclosure, the water inlet system of the laundry treating device includes a dispenser. The base of the dispenser is connected to the cover plate of the dispenser through the fastener, and the water storing part on the base is configured to define the water flow boundary. In a case that the base is connected to the cover plate, the water storing part and the cover plate collectively define a water flow channel. Meanwhile, under the fastening action of the fastener, the insertion part formed at the upper end of the water storing part fits tightly into the fitting slot formed on the cover plate, and sealing surfaces, which are in close contact with each other, are formed between the insertion part and the fitting slot, to guarantee the sealing performance of the water flow channel. Since the base is connected to the cover plate by means of locking through the fastener, and meanwhile the channel is sealed, the

cost can be reduced, and the production efficiency is relatively high compared to the welding method used in the related art.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The embodiments of the present disclosure will become apparent and more readily appreciated from the following description of the embodiments made with reference to the drawings, in which:
- (2) FIG. 1 is a schematic view of a dispenser according to an embodiment of the present disclosure;
- (3) FIG. 2 is a schematic view of a dispenser according to an embodiment of the present disclosure in another angle of view;
- (4) FIG. 3 is a schematic top view of a dispenser according to an embodiment of the present disclosure;
- (5) FIG. 4 is a schematic sectional view taken along a A-A direction in FIG. 3;
- (6) FIG. 5 is a schematic enlarged view of a portion B in FIG. 4;
- (7) FIG. 6 is a schematic view of an inner structure of a dispenser according to an embodiment of the present disclosure; and
- (8) FIG. 7 is a schematic view of a water inlet system according to an embodiment of the present disclosure.

LIST OF REFERENCE SYMBOLS

- (9) **100**: dispenser; **101**: first sealing surface, **102**: second sealing surface, **103**: third sealing surface, **104**: fourth sealing surface; **10**: base; **11**: water storing part, **111**: insertion part, **1111**: first inclined surface, **112**: spray chamber, **1121**: water outlet hole, **113**: first flow channel, **1131**: first water inlet section, **1132**: first water outlet section, **114**: second flow channel, **1141**: second water inlet section, **1142**: second water outlet section, **115**: third flow channel, **12**: receiving tray, **121**: water leakage hole, **13**: threaded hole, **14**: through hole; **20**: cover plate; **21**: fitting slot, **211**: narrowed section, **212**: protrusion, **213**: second inclined surface; **30**: fastener; **40**: snap assembly, **41**: snap, **42**: catch; **50**: first water inlet joint; **60**: second water inlet joint; **70**: first water outlet joint; **80**: second water outlet joint; **90**: hot water pipe joint; **200**: water inlet system; **210**: first control valve, **220**: second control valve, **230**: first detergent dispenser, **240**: second detergent dispenser, **260**: first hose, **270**: second hose, **280**: third control valve.

DETAILED DESCRIPTION OF THE DISCLOSURE

- (10) Exemplary embodiments of the present disclosure will be described in more detail below with reference to the accompanying drawings. Although the exemplary implementations of the present disclosure are shown in the accompanying drawings, it should be understood that the present disclosure can be implemented in various forms and should not be limited by the embodiments illustrated herein. These embodiments are provided to enable the present disclosure to be understood more thoroughly and to completely convey the scope of the present disclosure.
- (11) It should be understood that the terms used herein are intended only to describe some embodiments and are not limitation of the present disclosure. Unless the context clearly indicates otherwise, the singular forms “a/an”, “one”, and “the/said” used herein may also indicate the plural forms. The terms “comprise”, “include”, “contain” and “have” are inclusive, and thus indicate the presence of the features, steps, operations, elements, and/or components described, and they are not intended to (and do not) exclude the presence or addition of one or more other features, steps, operations, elements, components and/or groups thereof. The method steps, processes and operations described herein are not interpreted that they must be performed in the specific order as described or illustrated, unless the order is explicitly stated. It should also be understood that additional or alternative steps may be used.
- (12) Although the terms “first”, “second”, “third” and so on may be used to describe various

elements, components, regions, layers, and/or sections, these elements, components, regions, layers, and/or sections should not be limited by these terms. These terms are used merely to distinguish an element, component, region, layer, or section from another element, component, region, layer, or section. Unless otherwise indicated herein, terms such as “first” and “second” and other numerical terms do not imply order or sequence when used herein. Therefore, the first element, component, region, layer or section discussed below may be represented as a second element, component, region, layer or section without departing from the teaching of the exemplary implementations.

(13) For the convenience of description, spatial relative relation terms, such as “inner”, “outer”, “inside”, “outside”, “below”, “under”, “on”, “above”, may be used herein to describe a relationship of one element or feature and another element or feature as shown in the figures. The spatial relative relation terms include different orientations of the devices in use and operation other than the orientations shown in the figures. For example, if the device in the drawings is turned upside down, then elements described as “under”, or “below” other elements or features will be oriented to be “on” or “above” other elements or features. Therefore, the exemplary terms “under” may include both upper and lower orientations. The device may be otherwise oriented (rotated by 90 degrees or other orientations) and the spatial relative relation terms used herein are interpreted accordingly.

(14) As shown in FIG. 1 to FIG. 5, an embodiment of the present disclosure proposes a dispenser **100**. The dispenser **100** includes a base **10**, a cover plate **20** and a fastener **30**. In one embodiment, the base **10** includes a water storing part **11** configured to define a water flow boundary. An insertion part **111** is formed at an upper end of the water storing part **11**. A fitting slot **21** is formed on the cover plate **20**. The insertion part **111** is limited in the fitting slot **21**. The fastener **30** fastens the cover plate **20** to the base **10**, to enable sealing surfaces to be formed between the insertion part **111** and the fitting slot **21**, the sealing surfaces being in close contact with each other.

(15) According to the dispenser **100** in the embodiment of the present disclosure, the base **10** of the dispenser is connected to the cover plate **20** of the dispenser through the fastener **30**, and the water storing part **11** on the base **10** is configured to define the water flow boundary. In a case that the base **10** is connected to the cover plate **20**, the water storing part **11** and the cover plate **20** collectively define a water flow channel. Meanwhile, under the fastening action of the fastener **30**, the insertion part **111** formed at the upper end of the water storing part **11** fits tightly into the fitting slot **21** formed on the cover plate **20**, and sealing surfaces, which are in close contact with each other, are formed between the insertion part and the fitting slot, to guarantee the sealing performance of the water flow channel. Since the base **10** is connected to the cover plate **20** by means of locking through the fastener **30**, and meanwhile the channel is sealed, the cost can be reduced, and the production efficiency is relatively high compared to the welding method used in the related art.

(16) In some embodiments of the present disclosure, the fastener **30** is a screw. Correspondingly, the base **10** may be provided with a threaded hole **13**, and the cover plate **20** may be provided with a through hole **14**. The screw passes through the through hole **14**, and then is screwed into the threaded hole **13**, and to fasten the cover plate **20** to the base **10**. In some other embodiments of the present disclosure, the fastener **30** is a pin, which may be pinned through the cover plate **20** into the base **10**, and to fasten the cover plate **20** to the base **10**.

(17) In some embodiments of the present disclosure, the dispenser **100** further includes a snap assembly **40** arranged between the base **10** and the cover plate **20**. During connection of the cover plate **20** with the base **10** through the fastener **30**, the snap assembly is brought into a snapped state. In one embodiment, the snap assembly **40** may include a snap **41** and a catch **42**. The snap **41** is arranged on one of the base **10** and the cover plate **20**, and the catch **42** is arranged on the other of the base **10** and the cover plate **20**. In this embodiment, the snap assembly **40** is provided on the dispenser **100**, and the snap assembly **40** may also fasten the base **10** to the cover plate **20**. In a case

that the snap assembly **40** is provided, the number of fasteners **30** may be reduced. Since the snap assembly **40** may be brought into the snapped state automatically during connection of the fastener **30**, that is, no additional operation is required to bring the snap assembly **40** into the snapped state, the reduction of the number of the fasteners **30** may reduce the operation actions during the assembly of the cover plate **20** and the base **10**, which facilitates further improving the production efficiency.

(18) In some embodiments of the present disclosure, a bottom portion of the fitting slot **21** is provided with a narrowed section **211**. Under a fastening action of the fastener **30**, an end of the insertion part **111** abuts tightly against inner walls of the narrowed section **211** to form a first sealing surface **101** and a second sealing surface **102**. In this embodiment, since the bottom portion of the fitting slot **21** is provided with the narrowed section **211**, the bottom portion of the fitting slot **21** has a narrowing tendency in which a cross-sectional area of the bottom portion of the fitting slot **21** is gradually reduced. Thus, during the connection of the cover plate **20** with the base **10** through the fastener **30**, with gradually locking of the fastener **30**, two edges of the end of the insertion part **111** and two inner walls of the narrowed section **211** press against each other, and to enable the first sealing surface **101** and the second sealing surface **102** to be formed between the end of the insertion part **111** and the inner walls of the narrowed section **211**. The two sealing surfaces may provide a good sealing function on the water flow channel.

(19) Further, a protrusion **212** is formed at the bottom portion of the fitting slot **21**. Under the fastening action of the fastener **30**, the end of the insertion part **111** abuts tightly against the protrusion **212**, to form a third sealing surface **103**. In this embodiment, the protrusion **212** is formed at the bottom portion of the fitting slot **21**. During the connection of the cover plate **20** with the base **10** through the fastener **30**, with gradually locking of the fastener **30**, the protrusion **212** and the end of the insertion part **111** press against each other, and to enable the third sealing surface **103** to be formed between the end of the insertion part **111** and the protrusion **212**. Thus, the sealing performance of the water flow channel may be further improved, and the channel may be subjected to higher water pressure.

(20) Further, a first inclined surface **1111** is formed on the insertion part **111**, and a second inclined surface **213** is formed on an inner wall of the fitting slot **21**. Under the fastening action of the fastener **30**, the first inclined surface **1111** abuts tightly against the second inclined surface **213**, and to form a fourth sealing surface **104**. In this embodiment, the first inclined surface **1111** is formed on the insertion part **111**, and the second inclined surface **213** is formed on the inner wall of the fitting slot **21**. Thus, during the connection of the cover plate **20** with the base **10** through the fastener **30**, with gradually locking of the fastener **30**, the first inclined surface **1111** and the second inclined surface **213** press against each other, and to form the fourth sealing surface **104** between the insertion part **111** and the inner wall of the fitting slot **211**. Thus, the sealing performance of the water flow channel may be further improved, and the channel may be subjected to higher water pressure.

(21) Further, the base **10** and the cover plate **20** are both plastic components, and the base **10** and the cover plate **20** both have a level of deformability. Thus, during the connection of the cover plate **20** with the base **10** through the fastener **30**, several contact portions of the insertion part **111** with the fitting slot **21** press against each other to be deformed to a certain extent, and the contact portions of the insertion part **111** with the fitting slot **21** are attached to each other more tightly, to form the sealing surfaces with good sealing performance.

(22) In some embodiments of the present disclosure, the water storing part **11** includes a flow channel assembly, and a spray chamber **112** connected to the flow channel assembly. Water outlet holes **1121** are formed at a bottom part of the spray chamber **112**. In a case that the base **10** is connected to the cover plate **20**, the spray chamber **112** and the cover plate **20** form a spray box. The spray box is configured to allow the tap water to be sprayed into the washing drum of the laundry treating device. Spraying the water provides an advantage that the laundry placed in the

washing drum may be wetted thoroughly and more evenly before the laundry is washed. In addition, during rinsing of the laundry in the laundry treating device, the laundry may be rinsed in a spray manner, which may not only save water, but also achieve better rinsing effects.

(23) In some embodiments of the present disclosure, as shown in FIG. 6 and FIG. 7, the flow channel assembly includes an X-shaped flow channel. The X-shaped flow channel includes a first flow channel **113** and a second flow channel **114** intersecting with the first flow channel. The first flow channel **113** includes a first water inlet section **1131** and a first water outlet section **1132**, and the second flow channel **114** includes a second water inlet section **1141** and a second water outlet section **1142**. The flow channel assembly further includes a third flow channel **115**. The third flow channel **115** is arranged between the first water outlet section **1132** and the second water outlet section **1142**. The third flow channel **115** is in communication with the X-shaped flow channel. An inlet of the third flow channel **115** is arranged at the intersection of the first flow channel **113** with the second flow channel **114**. The spray chamber **112** is in communication with the third flow channel **115**. When the dispenser **100** is applied to the water inlet system of the laundry treating device, each of the first flow channel **113** and the second flow channel **114** of the X-shaped flow channel may be connected to a respective one of the detergent dispensers, and a control valve may be installed at each of the inlets of the first flow channel **113** and the second flow channel **114**. For example, the first flow channel **113** is connected to a first detergent dispenser **230**, and the second flow channel **114** is connected to a second detergent dispenser **240**. The first detergent dispenser **230** and the second detergent dispenser **240** are configured to receive different washing agents (for example, detergent, softener, etc.). Meanwhile, a first control valve **210** is installed at the inlet of the first flow channel **113**, and a second control valve **220** is installed at the inlet of the second flow channel **114**.

(24) In the process of introducing water into the laundry treating device, in a case that only the first control valve **210** is opened, the tap water enters into the first flow channel **113** through the first control valve **210**. When the water flow reaches the intersection of the X-shaped flow channel along the first flow channel **113**, most part of the water flow will flow in its original flow direction and enter the first water outlet section **1132**, and another part of the water flow will enter into the third flow channel **115**. As such, most part of the water flow will enter the first detergent dispenser **230**, to flush the washing agent in the first detergent dispenser **230** into the washing drum of the laundry treating device. Meanwhile, another part of the water flow will enter into the spray chamber **112**, and is eventually sprayed by the spray chamber **112** into the washing drum. Similarly, in a case that only the second control valve **220** is opened, the tap water enters into the second flow channel **114** through the second control valve **220**. Most part of the water flow will eventually enter into the second detergent dispenser **240** through the second water outlet section **1142** to flush the washing agent in the second detergent dispenser **240** into the washing drum, and another part of the water flow will be sprayed via the spray chamber **112** into the washing drum. Moreover, in a case that the first control valve **210** and the second control valve **220** are simultaneously opened, the tap water simultaneously enters into the first flow channel **113** and the second flow channel **114**. After these two water flows are conflated at the intersection of the X-shaped flow channel, they enter into the spray chamber **112** and are eventually sprayed into the washing drum.

(25) It can be seen that with two control valves, i.e., the first control valve **210** and the second control valve **220**, a variety of water inlet modes, such as flush of the first detergent dispenser **230**, flush of the second detergent dispenser **240**, spraying of the water, etc., can be achieved. Compared with the existing water inlet system, fewer control valves are used, and the structure is simplified, and the cost is reduced.

(26) It can be understood that the first detergent dispenser **230** and the second detergent dispenser **240** are configured to receive different washing agents. In a common implementation, the first detergent dispenser **230** is configured to receive detergent or washing power, and the second detergent dispenser **240** is configured to receive softener. During rinsing of the laundry, addition of

the softener can reduce the coefficient of friction between the fibers, and restore the inherent smoothness, extension and compression properties of the fibers, allowing the laundry to become soft, fluffy and elastic.

(27) In some embodiments of the present disclosure, the dispenser **100** further includes a first water inlet joint **50** and a second water inlet joint **60**. The first water inlet joint **50** is arranged at an inlet of the first flow channel **113**, and the second water inlet joint **60** is arranged at an inlet of the second flow channel **114**. The first water inlet joint **50** is configured to connect the inlet end of the first flow channel **113** to an external pipeline. For example, in a specific example, the first control valve **210** is connected to the first water inlet joint **50**, to achieve the connection between the first control valve **210** and the inlet of the first flow channel **113**, to enable the first control valve **210** to control the opening and closing of the first flow channel **113**. Similarly, the second water inlet joint **60** is configured to connect the inlet end of the second flow channel **114** to an external pipeline. In one embodiment, the second control valve **220** is connected to the second water inlet joint **60**, to achieve the connection between the second control valve **220** and the inlet of the second flow channel **114**, to enable the second control valve **220** to control the opening and closing of the second flow channel **114**.

(28) In one embodiment, the inlet ends of the first water inlet joint **50** and the second water inlet joint **60** may be provided with internal threads or external threads, and the first water inlet joint **50** and the second water inlet joint **60** may be connected to the external pipeline by means of threaded connection. Such a connection method is easier to operate, and facilitates improvement of the assembly efficiency.

(29) It can be understood that the first water inlet joint **50** and the second water inlet joint **60** may be arranged on the base **10**, or may be arranged on the cover plate **20**. The specific positions thereof may be selected according to actual needs.

(30) In some embodiments of the present disclosure, the dispenser **100** further includes a first water outlet joint **70** and a second water outlet joint **80**. The first water outlet joint **70** is arranged at an outlet of the first flow channel **113**, and the second water outlet joint **80** is arranged at an outlet of the second flow channel **114**. The first water outlet joint **70** is configured to connect the outlet end of the first flow channel **113** and an external pipeline. In one embodiment, the first water outlet joint **70** is connected to a hose in communication with the first detergent dispenser **230**, and the connection between the first flow channel **113** and the first detergent dispenser **230** can be achieved. Similarly, the second water outlet joint **80** is configured to connect the outlet end of the second flow channel **114** to an external pipeline. In one embodiment, the second water outlet joint **80** is connected to a hose in communication with the second detergent dispenser **240**, and the connection between the second flow channel **114** and the second detergent dispenser **240** can be achieved.

(31) In one embodiment, the outlet ends of the first water outlet joint **70** and the second water outlet joint **80** may be provided with internal threads or external threads, and the first water outlet joint **70** and the second water outlet joint **80** may be connected to the external pipeline by means of threaded connection. Such a connection method is easier to operate, and facilitates improvement of the assembly efficiency.

(32) It can be understood that the first water outlet joint **70** and the second water outlet joint **80** may be arranged on the base **10**, or may be arranged on the cover plate **20**. The specific positions thereof may be selected according to actual needs.

(33) In some embodiments of the present disclosure, the dispenser **100** further includes a hot water pipe joint **90** arranged on the cover plate **20**. The hot water pipe joint **90** is in communication with the spray chamber **112**. The hot water pipe joint **90** may be connected to an external hot water source via a water pipe, and the hot water can be introduced into the laundry treating device, and the water path for the hot water and the water path for the tap water (cold water) are independent of each other. In one embodiment, when the hot water is introduced, the hot water supplied by the

external hot water source directly enters into the spray chamber **112** via the hot water pipe joint **90**, and then is sprayed into the washing drum of the laundry treating device. In this embodiment, the hot water may be introduced into the laundry treating device. By controlling the ratio of the hot water to the tap water entering into the washing drum, the temperature of the washing water may be adjusted, to allow the temperature of the washing water to be in a better temperature range, to improve the cleaning effect on the laundry.

(34) In some embodiments of the present disclosure, the base **10** further includes a receiving tray **12**. The receiving tray **12** is arranged around the flow channel assembly. The receiving tray **12** is provided with water leakage holes **121**. In a case of water leakage from the flow channel assembly due to poor sealing or excessive water pressure, the receiving tray **12** may be used to receive the water leaked from the flow channel assembly, to avoid the leaked water from scattering onto other components of the laundry treating device, which adversely affect the other components. In addition, the water leakage holes on the receiving tray **12** may be further connected to a guide structure (for example, a water pipe, a drainage trough, etc.). The guide structure is in communication with the washing drum of the laundry treating device, to enable the water in the receiving tray **12** to be further guided into the washing drum.

(35) Further, when the dispenser **100** is installed in the laundry treating device, the receiving tray **12** may be arranged at an angle to the horizontal plane, that is, the receiving tray **12** may be slightly inclined. In addition, the water leakage holes **121** are arranged at a low level. In this way, this facilitates allowing the water in the receiving tray **12** to be collected towards the water leakage holes **121**, and eventually flow into the washing drum of the laundry treating device via the water leakage holes **121** and the guide structure.

(36) An embodiment of the present disclosure proposes a water inlet system **200**. The water inlet system **200** includes the dispenser **100** in any one of the above embodiments.

(37) According to the water inlet system **200** in the embodiment of the present disclosure, the base **10** of the dispenser **100** of the water inlet system is connected to the cover plate **20** of the dispenser **100** of the water inlet system through the fastener **30**, and the water storing part **11** on the base **10** is configured to define the water flow boundary. In a case that the base **10** is connected to the cover plate **20**, the water storing part **11** and the cover plate **20** collectively define a water flow channel. Meanwhile, under the fastening action of the fastener **30**, the insertion part **111** formed at the upper end of the water storing part **11** fits tightly into the fitting slot **21** formed on the cover plate **20**, and sealing surfaces, which are in close contact with each other, are formed between the insertion part and the fitting slot, to guarantee the sealing performance of the water flow channel. Since the base **10** is connected to the cover plate **20** by means of locking through the fastener **30**, and meanwhile the channel is sealed, the cost can be reduced, and the production efficiency is relatively high compared to the welding method used in the related art.

(38) In some embodiments of the present disclosure, the flow channel assembly includes an X-shaped flow channel. The X-shaped flow channel includes a first flow channel **113** and a second flow channel **114** intersecting with the first flow channel. The first flow channel **113** includes a first water inlet section **1131** and a first water outlet section **1132**, and the second flow channel **114** includes a second water inlet section **1141** and a second water outlet section **1142**. The flow channel assembly further includes a third flow channel **115**. The third flow channel **115** is arranged between the first water outlet section **1132** and the second water outlet section **1142**. The third flow channel **115** is in communication with the X-shaped flow channel. An inlet of the third flow channel **115** is arranged at the intersection of the first flow channel **113** with the second flow channel **114**. The spray chamber **112** is in communication with the third flow channel **115**. The water inlet system **200** further includes a first control valve **210**, a second control valve **220**, a first detergent dispenser **230** and a second detergent dispenser **240**. The first control valve **210** is installed at the inlet of the first flow channel **113**, and the second control valve **220** is installed at the inlet of the second flow channel **114**. The first detergent dispenser **230** is connected to the first

flow channel **113**, and the second detergent dispenser **240** is connected to the second flow channel **114**.

(39) In this embodiment, the first flow channel **113** in the X-shaped flow channel is connected to the first detergent dispenser **230**, and the second flow channel **114** is connected to the second detergent dispenser **240**. The first detergent dispenser **230** and the second detergent dispenser **240** are configured to receive different washing agents (for example, detergent, softener, etc.). Meanwhile, the first control valve **210** is installed at the inlet of the first flow channel **113**, and the second control valve **220** is installed at the inlet of the second flow channel **114**. In the process of introducing water into the laundry treating device, in a case that only the first control valve **210** is opened, the tap water enters into the first flow channel **113** through the first control valve **210**. When the water flow reaches the intersection of the X-shaped flow channel along the first flow channel **113**, most part of the water flow will flow in its original flow direction and enter the first water outlet section **1132**, and another part of the water flow will enter into the third flow channel **115**. As such, most part of the water flow will enter the first detergent dispenser **230**, to flush the washing agent in the first detergent dispenser **230** into the washing drum of the laundry treating device. Meanwhile, another part of the water flow will enter into the spray chamber **112**, and is eventually sprayed by the spray chamber **112** into the washing drum. Similarly, in a case that only the second control valve **220** is opened, the tap water enters into the second flow channel **114** through the second control valve **220**. Most part of the water flow will eventually enter into the second detergent dispenser **240** through the second water outlet section **1142** to flush the washing agent in the second detergent dispenser **240** into the washing drum, and another part of the water flow will be sprayed via the spray chamber **112** into the washing drum. Moreover, in a case that the first control valve **210** and the second control valve **220** are simultaneously opened, the tap water simultaneously enters into the first flow channel **113** and the second flow channel **114**. After these two water flows are conflated at the intersection of the X-shaped flow channel, they enter into the spray chamber **112** and are eventually sprayed into the washing drum. It can be seen that with two control valves, i.e., the first control valve **210** and the second control valve **220**, a variety of water inlet modes, such as flush of the first detergent dispenser **230**, flush of the second detergent dispenser **240**, spraying of the water, etc., can be achieved. Compared with the existing water inlet system, fewer control valves are used, and the structure is simplified, and the cost is reduced.

(40) In some embodiments of the present disclosure, the water inlet system **200** further includes a first hose **260** and a second hose **270**. An end of the first hose **260** is connected to the outlet of the first flow channel **113**, and another end of the first hose **260** is connected to the first detergent dispenser **230**. An end of the second hose **270** is connected to the outlet of the second flow channel **114**, and another end of the second hose **270** is connected to the second detergent dispenser **240**. In this embodiment, the first detergent dispenser **230** is connected to the first flow channel **113** through the first hose **260**, and the second detergent dispenser **240** is connected to the second flow channel **114** through the second hose **270**. In this way, the first detergent dispenser **230** and the second detergent dispenser **240** may be arranged at suitable positions in the laundry treating device according to design needs, without being limited to positions adjacent to the dispenser **100**, to provide a larger design freedom for the structural design of the laundry treating device. In addition, the first hose **260** and the second hose **270** are provided in such a way that the flow path of the water flow becomes longer, and the water flow flowing towards the first detergent dispenser **230** or the second detergent dispenser **240** is subjected to greater resistance. In this way, this facilitates allowing the water flow flowing towards the first detergent dispenser **230** or the second detergent dispenser **240** to be partly diverted at the intersection of the X-shaped flow channel **20** to the third flow channel **30**, and to ensure that the water flow enters into the washing drum of the laundry treating device through the spray box **40** in all water inflow modes, to maintain the effect of spraying water on the laundry.

(41) In some embodiments of the present disclosure, the water inlet system **200** further includes a

third control valve **280**. The dispenser **100** further includes a hot water pipe joint **90** arranged on the spray box **40**. The hot water pipe joint **90** is connected to an inlet of the third control valve **280**. In this embodiment, the water inlet system **200** may further introduce the hot water into the laundry treating device through the third control valve **280** and the hot water pipe joint **90**. In one embodiment, the third control valve **280** is configured to control the opening and closing of the water path for the hot water. In a case that the third control valve **280** is opened, the hot water supplied by the external hot water source sequentially passes through the third control valve **280** and the hot water pipe joint **90**, and then enters into the spray chamber **112**, and then is sprayed into the washing drum of the laundry treating device. In the water inlet system **200** in this embodiment, by controlling the ratio of the hot water to the tap water entering into the washing drum, the temperature of the washing water may be adjusted, to allow the temperature of the washing water to be in a better temperature range, to improve the cleaning effect on the laundry.

(42) An embodiment of the present disclosure proposes a laundry treating device (such as a washing machine, a washing and drying machine, etc.), which includes the water inlet system in any one of the above embodiments.

(43) According to the laundry treating device in the embodiment of the present disclosure, the water inlet system of the laundry treating device includes a dispenser **100**. The base **10** of the dispenser **100** is connected to the cover plate **20** of the dispenser **100** through the fastener **30**, and the water storing part **11** on the base **10** is configured to define the water flow boundary. In a case that the base **10** is connected to the cover plate **20**, the water storing part **11** and the cover plate **20** collectively define a water flow channel. Meanwhile, under the fastening action of the fastener **30**, the insertion part **111** formed at the upper end of the water storing part **11** fits tightly into the fitting slot **21** formed on the cover plate **20**, and sealing surfaces, which are in close contact with each other, are formed between the insertion part and the fitting slot, to guarantee the sealing performance of the water flow channel. Since the base **10** is connected to the cover plate **20** by means of locking through the fastener **30**, and meanwhile the channel is sealed, the cost can be reduced, and the production efficiency is relatively high compared to the welding method used in the related art.

Claims

1. A dispenser, comprising: a base, wherein the base comprises a water storing part configured to define a water flow boundary, an insertion part being formed at an upper end of the water storing part; a cover plate, wherein a fitting slot is formed on the cover plate, the insertion part being limited in the fitting slot; and a fastener, wherein the fastener fastens the cover plate to the base, and wherein a plurality of sealing surfaces are formed between the insertion part and the fitting slot, the plurality of sealing surfaces being in close contact with each other; wherein a bottom portion of the fitting slot is provided with a narrowed section, wherein under a fastening action of the fastener, an end of the insertion part abuts tightly against inner walls of the narrowed section to form a first sealing surface and a second sealing surface, and the second sealing surface is parallel and opposite to the first sealing surface.
2. The dispenser according to claim 1, wherein a protrusion is formed at the bottom portion of the fitting slot, wherein under the fastening action of the fastener, the end of the insertion part abuts tightly against the protrusion to form a third sealing surface, and a plane of the third sealing surface is perpendicular to planes of the first and the second sealing surfaces respectively.
3. The dispenser according to claim 2, wherein a first inclined surface is formed on the insertion part, and a second inclined surface is formed on an inner wall of the fitting slot, wherein under the fastening action of the fastener, the first inclined surface abuts tightly against the second inclined surface to form a fourth sealing surface.
4. The dispenser according to claim 1, wherein the water storing part comprises a flow channel

assembly, and a spray chamber connected to the flow channel assembly, wherein a plurality of water outlet holes are formed at a bottom part of the spray chamber.

5. The dispenser according to claim 4, wherein the flow channel assembly comprises: an X-shaped flow channel, wherein the X-shaped flow channel comprises a first flow channel and a second flow channel intersecting with the first flow channel, wherein the first flow channel comprises a first water inlet section and a first water outlet section, and the second flow channel comprises a second water inlet section and a second water outlet section; and a third flow channel, wherein the third flow channel is arranged between the first water outlet section and the second water outlet section, the third flow channel is in communication with the X-shaped flow channel, an inlet of the third flow channel is arranged at an intersection of the first flow channel with the second flow channel, and the spray chamber is in communication with the third flow channel.

6. The dispenser according to claim 5, further comprising: a first water inlet joint arranged at an inlet of the first flow channel; a second water inlet joint arranged at an inlet of the second flow channel; a first water outlet joint arranged at an outlet of the first flow channel; and a second water outlet joint arranged at an outlet of the second flow channel.

7. The dispenser according to claim 5, wherein the dispenser further comprises a hot water pipe joint arranged on the cover plate, wherein the hot water pipe joint is in communication with the spray chamber.

8. The dispenser according to claim 4, wherein the base further comprises a receiving tray, wherein the receiving tray is arranged around the flow channel assembly, and the receiving tray is provided with a plurality of water leakage holes.

9. The dispenser according to claim 1, wherein the dispenser further comprises a snap assembly arranged between the base and the cover plate.

10. A water inlet system, comprising: a dispenser, comprising: a base, wherein the base comprises a water storing part configured to define a water flow boundary, an insertion part being formed at an upper end of the water storing part; a cover plate, wherein a fitting slot is formed on the cover plate, the insertion part being limited in the fitting slot; and a fastener, wherein the fastener fastens the cover plate to the base to enable a plurality of sealing surfaces to be formed between the insertion part and the fitting slot, the plurality of sealing surfaces being in close contact with each other; wherein a bottom portion of the fitting slot is provided with a narrowed section, wherein under a fastening action of the fastener, an end of the insertion part abuts tightly against inner walls of the narrowed section to form a first sealing surface and a second sealing surface, and the second sealing surface is parallel and opposite to the first sealing surface.

11. A laundry treating device, comprising: a water inlet system, comprising: a dispenser, comprising: a base, wherein the base comprises a water storing part configured to define a water flow boundary, an insertion part being formed at an upper end of the water storing part; a cover plate, wherein a fitting slot is formed on the cover plate, the insertion part being limited in the fitting slot; and a fastener, wherein the fastener fastens the cover plate to the base to enable a plurality of sealing surfaces to be formed between the insertion part and the fitting slot, the plurality of sealing surfaces being in close contact with each other; wherein a bottom portion of the fitting slot is provided with a narrowed section, wherein under a fastening action of the fastener, an end of the insertion part abuts tightly against inner walls of the narrowed section to form a first sealing surface and a second sealing surface, and the second sealing surface is parallel and opposite to the first sealing surface.
